

FPT UNIVERSITY — HO CHI MINH CITY
Faculty of Information Technology

DSP501

Digital Signal and Image Processing

FINAL ASSIGNMENT

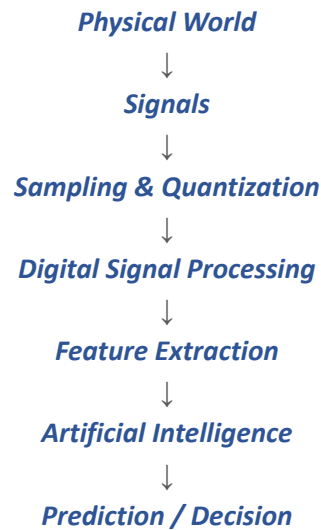
A Semester-long DSP–AI Research Project

Group Project • 3–4 students

Project Handbook • OBE • Research-Based Learning (RBL) • AI-Native Learning
Deliverables: IEEE Report + Reproducible Package + Presentation

The Big Picture — From Signals to Intelligence

The figure below shows where this project fits within the complete intelligent-system pipeline. DSP is the bridge that converts raw physical signals into meaningful representations on which AI makes predictions and decisions.



This assignment spans the **Sampling & Quantization** → **Feature Extraction** → **AI** stages: students design the DSP front-end and integrate it with an AI model to solve a real signal-processing problem.

1. Introduction

Purpose

The Final Assignment is a semester-long group project that accompanies the entire course. Students will investigate a real-world signal processing problem by integrating Digital Signal Processing (DSP) and Artificial Intelligence (AI). The project follows the principles of Research-Based Learning (RBL), where students progressively develop their solution through problem formulation, literature review, implementation, experimentation, and critical analysis.

Rather than completing the project only at the end of the course, students are expected to continuously refine their work as new DSP concepts are introduced throughout the semester. The assignment is problem-driven, not topic-driven: for example, the objective is not “Speech Emotion Recognition” in general, but a specific, well-defined problem such as “Robust Speech Emotion Recognition under Noisy Environments.”

How to Succeed in This Project

The following practical advice reflects what distinguishes strong projects from weak ones. It sets expectations only and introduces no additional assessment requirements.

- Choose a focused research problem — a narrow, well-defined problem is easier to investigate rigorously than a broad topic.
- Read the literature before implementation — understanding existing methods prevents reinventing weak baselines and clarifies the research gap.
- Design the DSP pipeline before selecting AI models — the signal representation often matters more than the choice of classifier.

- Conduct experiments continuously rather than at the end — iterate throughout the semester as new DSP concepts are introduced.
- Explain why the results occur — link findings back to DSP theory and the research questions, rather than only reporting performance numbers.
- Maintain reproducibility throughout — fix random seeds, record versions, and keep the run scripts working as the project evolves, not as an afterthought.

2. Learning Outcomes

Upon successful completion of the project, students will be able to:

- formulate a real-world signal processing problem;
- conduct a literature review and identify research gaps;
- design and justify DSP methodologies;
- implement DSP algorithms for signal preprocessing and feature extraction;
- integrate DSP with AI/ML models;
- design scientifically sound experiments;
- critically analyze experimental results;
- communicate research findings professionally.

Alignment with Course Learning Outcomes (CLOs)

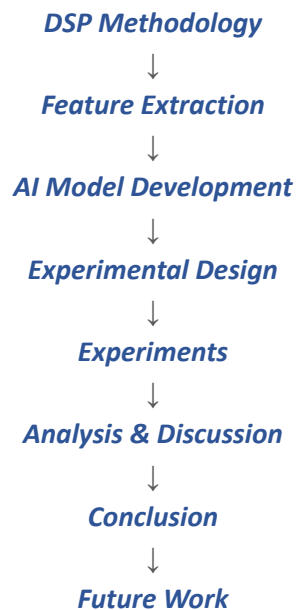
The project outcomes map to the course learning outcomes as follows (indicative alignment only):

Project Outcome	Related CLO
Problem formulation	CLO1
DSP methodology	CLO2
Frequency-domain analysis	CLO3
DSP–AI integration	CLO4
Experimental evaluation	CLO5

3. Project Workflow

Every project shall follow the research workflow below. This mirrors the process used when writing a scientific paper.





4. Project Scope

Each group (**3–4 students**) shall complete the following five phases across the semester.

Phase 1 — Problem Definition

- Select a real-world signal processing problem (specific and well-defined, with an associated research gap).
- Define 1–2 research objectives — what the project aims to achieve.
- Formulate 2–4 research questions that follow directly from the objectives.
- State one or more testable research hypotheses (if applicable), or a research proposition for exploratory projects.

Deliverables: *Research Problem; Research Objectives; Research Questions; Research Hypothesis or Proposition.*

Phase 2 — Literature Review

Conduct a literature review of 8–12 recent publications (2021–2026). The review should:

- compare DSP techniques across papers;
- compare AI approaches (models, datasets, findings) rather than listing summaries;
- identify the research gap the project addresses;
- summarize the expected research contributions.

Composition: at least **3 journal papers** and **3 conference papers**; at least **2 survey/review papers** are **recommended** (reading surveys first helps orient the more specialized reading). A **Literature Matrix is required** (see Appendix A).

Deliverables: *Literature Matrix; Research Gap; Expected Contributions.*

Phase 3 — DSP Methodology

Design a complete DSP pipeline including signal acquisition, preprocessing, digital filtering, frequency analysis, and feature extraction. All design decisions must be theoretically justified.

Deliverables: DSP pipeline design; justified preprocessing and filtering choices; feature-extraction plan.

Phase 4 — AI Methodology

Develop AI/ML models for classification, recognition, detection, or prediction. Model selection and hyperparameters must be justified and reported (learning rate, batch size, number of epochs, optimizer, etc.).

Deliverables: Model architecture and rationale; hyperparameter configuration; trained model.

Phase 5 — Experimental Evaluation

Experiments should answer the research questions, not merely report performance. Each project must include a baseline comparison, an ablation study, quantitative evaluation, error analysis, and discussion.

Deliverables: Baseline comparison; ablation study; quantitative results; error analysis; discussion.

5. Research Questions & Hypotheses

Each project must formulate **2–4 research questions (RQs)** that are specific, measurable, and grounded in the literature review, plus **one or more testable hypotheses where applicable**. For exploratory projects in which a formal hypothesis is not appropriate, a **research proposition** may be stated instead. Illustrative examples:

Example research questions

- **RQ1.** How does digital filtering affect signal quality and downstream classification accuracy?
- **RQ2.** Which feature extraction technique (FFT, STFT, Wavelet, MFCC) performs best for the selected task?
- **RQ3.** Does AI significantly improve performance over classical DSP-only approaches?
- **RQ4.** What factors influence the robustness of the proposed model?

Example hypotheses (or propositions)

- **H1.** MFCC combined with STFT features achieves higher recognition accuracy than MFCC alone.
- **H2.** Band-pass filtering significantly improves ECG classification performance.

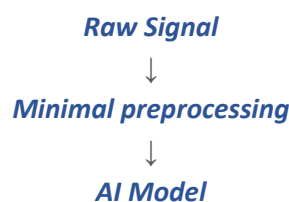
Reference — FINER framework

A useful guide for formulating good research questions (Feasible, Interesting, Novel, Ethical, Relevant):
<https://scientific-publishing.webshop.elsevier.com/research-process/finer-research-framework/>

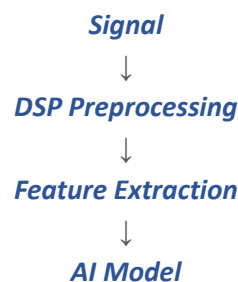
6. Mandatory DSP Pipeline (Two-System Comparison)

Every project must implement and **compare two systems** to experimentally demonstrate the impact of DSP preprocessing on AI performance.

Pipeline A — Baseline (minimal DSP)



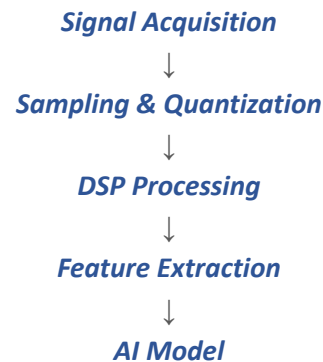
Pipeline B — Proposed (full DSP)



Standard pipeline reference: Signal Input → Preprocessing (DSP) → Feature Extraction → AI/ML Model → Output Classification. Each stage must be clearly explained, implemented, and justified with respect to the research questions.

7. Minimum Technical Requirements

Signal acquisition and sampling are **prerequisite stages**, not DSP techniques. The overall workflow is:



Within the DSP Processing and Feature Extraction stages, each project must include **at least three DSP techniques** selected from the table below. Projects relying on only one DSP technique (e.g., MFCC only) **will not satisfy the minimum requirements**.

Category	Techniques
Frequency analysis	FFT, STFT, Wavelet Transform
Filtering	Digital filtering (FIR/IIR): bandpass, notch, low-pass
Spectral / power	Power Spectral Density (PSD), band-power features
Cepstral	MFCC
Statistical / information	Statistical features, Entropy features

Visualization is required: plot raw signals, spectrograms, and extracted features (spectrogram or frequency-domain visualization).

8. AI / Machine Learning Requirements

- Design and evaluate at least one AI/ML model (SVM, Random Forest, CNN, LSTM, GRU, etc.).
- Evaluate using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- Baseline comparison is required: proposed approach vs. at least one baseline (e.g., DSP-only vs. DSP + AI, or FFT vs. STFT vs. Wavelet features).
- Hyperparameter tuning must be reported (learning rate, batch size, epochs, optimizer, etc.).

Recommended (not mandatory)

Model explainability (e.g., SHAP, LIME, Grad-CAM, or other appropriate techniques) is encouraged for interpreting model decisions and understanding which signal features drive predictions — especially valuable at master's level.

9. Experimental Design

Experiments must be explicitly designed to answer the research questions. Each group must specify:

- **Independent variables:** e.g., feature type, filter configuration, model architecture.
- **Dependent variables:** e.g., accuracy, F1-score, inference time.
- **Evaluation metrics & protocol:** train/test split or cross-validation, random seeds, number of runs.
- **Controlled conditions:** ensure a fair comparison between baselines and the proposed method.

Required & encouraged components

- Ablation study (required): demonstrate the contribution of each pipeline component. If the method uses filtering + MFCC + CNN, also evaluate without filtering, without MFCC, and FFT-features-only.
- Statistical significance (encouraged): paired t-test, Wilcoxon signed-rank test, or 95% confidence intervals over repeated runs.
- Error analysis (required): analyze misclassified samples, failure cases, noisy signals, and difficult classes to explain where and why the model fails.

10. Implementation & Reproducibility

- Programming environment: Python or MATLAB.
- Suggested Python libraries: NumPy, SciPy, Librosa, Scikit-learn, TensorFlow/Keras, Matplotlib.
- Use public datasets (DEAP, RAVDESS, EMO-DB, PhysioNet, etc.) or self-recorded data (with ethical consent).
- Code must be runnable, well-commented, and well-structured.

Reproducibility (required)

Provide everything needed for another person to re-run the experiments: fixed random seeds, software versions, hardware configuration, a complete dependency list (requirements.txt), the trained model, the experiment configuration, and step-by-step reproduction instructions. Reproducibility is met only if the experiments can be re-run from the submitted package.

11. Suggested Project Domains

Examples include (custom topics are welcome with instructor approval):

- **Speech / Audio:** Speech Emotion Recognition, Speaker Identification, Environmental Sound Classification
- **Biomedical:** ECG Arrhythmia Detection, EEG Sleep Stage Classification
- **Industrial / Structural:** Machinery Fault Detection, Structural Vibration Analysis

12. Project Timeline

Lecture	Milestone
L1	Course & project introduction
L2	Assignment announcement, team formation
L3	Topic registration
L4	Research Questions + Literature Review checkpoint
L5	Research proposal
L6	Methodology consultation
L7	Progress review
L8	Experimental refinement
L9	Final report, source code & slides submission
L10	Final presentation & demonstration

All milestones before Lecture 9 are **formative feedback only** — they guide progress and do not carry separate grades.

13. Deliverables

Required

- Technical Report — 8–10 pages (excluding references and appendices), single-column IEEE format. Overleaf template: <https://www.overleaf.com/read/zrmhstndnmwd#c703cb>
- Literature Matrix (Paper | DSP | AI | Dataset | Findings | Research Gap | Limitations) — may be placed in the report appendix.
- Reproducible Research Package — a self-contained package another person can download and run immediately: README.md, requirements.txt, run.py, dataset link, trained model, and experiment configuration.
- Presentation Slides — 10–12 minutes per group.

Optional

- Demonstration Video (3–5 minutes).
- Research Poster.
- One-page Extended Abstract — recommended for developing the project into a conference submission.

14. Report Structure (Required)

The report shall include the following sections. The Methodology section must **begin with a Research Framework diagram** (Research Problem → Dataset → DSP → Feature Extraction → Model → Evaluation), then detail Dataset, DSP Preprocessing, Feature Extraction, and AI/ML Model Design.

Following IEEE paper conventions, the report begins with an **Abstract** and **Keywords** before the Introduction.

1. Abstract
2. Keywords
3. Introduction
4. Research Problem
5. Research Objectives
6. Research Questions
7. Literature Review
8. Research Gap
9. Expected Contributions
10. Methodology
11. Experimental Design
12. Experimental Results
13. Error Analysis (misclassified samples, failure cases, difficult classes)
14. Discussion
15. Threats to Validity
16. Conclusion
17. Future Work
18. Ethics Statement
19. AI Declaration (Appendix)

Discussion requirements

The Discussion must go beyond accuracy figures. Analyze WHY the results occurred (link findings to DSP theory and the RQs); the STRENGTHS of the approach relative to baselines and the literature; the WEAKNESSES and limitations of method, data, and evaluation; and the THREATS TO VALIDITY (internal, external, construct) and how they were mitigated.

15. Evaluation Criteria

The rubric evaluates the research process as well as the final system.

Criterion	Weight
Research Problem, Objectives & Literature Review	15%
DSP Methodology	20%
AI/ML Methodology	20%
Experimental Design, Baseline Comparison & Evaluation	20%
Discussion, Research Findings & Threats to Validity	10%

Criterion	Weight
Report & Presentation	10%
Research Contribution & Reproducibility	5%

Report & Presentation evaluates communication quality (clarity, structure, and delivery); **Research Contribution & Reproducibility** evaluates originality, engineering contribution, and the reproducibility of the submitted package.

16. Research Ethics

Because DSP501 projects involve human speech and biomedical signals (ECG, EEG), all projects must comply with basic research ethics. Students must:

- respect dataset licenses and cite datasets according to their terms of use;
- anonymize all human speech and biomedical data; remove personally identifiable information;
- obtain informed consent for any self-recorded data, and state the consent procedure in the report;
- acknowledge external code and datasets;
- ensure reproducibility of experiments.

An Ethics Statement must be included in the report.

17. AI Declaration (Required)

Generative AI tools (e.g., ChatGPT, Gemini, Claude, GitHub Copilot) may be used to support learning and research. However, students remain fully responsible for the originality, correctness, and integrity of all submitted work. Each group must include an AI Declaration in the appendix of the report, stating:

- AI tool(s) used;
- purpose of use;
- stages where AI was used;
- human verification and modifications;
- responsibility statement.

Acceptable uses

- brainstorming research ideas;
- improving academic writing;
- summarizing literature;
- explaining DSP concepts;
- debugging code;
- improving figures and presentations.

Unacceptable uses

- fabricated experimental results;
- fabricated references;
- fabricated datasets;
- submitting AI-generated reports without verification;
- copying AI outputs without understanding.

18. Individual AI Reflection (Required)

Each student shall submit an **individual reflection (300–500 words)** describing:

- How AI supported your project.
- Which AI tools were used.
- What limitations or errors you encountered.
- How AI-generated outputs were verified.
- What engineering or research decisions required your own judgment.
- How you would improve your AI usage in future projects.

Guiding questions

The following questions are intended to guide the reflection without restricting the answers:

- Which AI tools did you use?
- At which project stages was AI most helpful?
- What errors or hallucinations did AI produce?
- How did you verify AI-generated outputs?
- What engineering decisions required your own judgment?

The reflection is intended to promote responsible, transparent, and reflective use of generative AI.

19. Academic Integrity

Students are expected to uphold the highest standards of academic integrity. The following are strictly prohibited:

- plagiarism;
- fabrication or falsification of data;
- manipulated experimental results;
- undisclosed AI-generated content;
- uncredited reuse of code or datasets.

Consequences

Violations may result in a zero score for the assignment and further disciplinary action according to university regulations.

20. Registration

Topic registration form: <https://forms.gle/KNAdp35hSN25MAJVA>

Appendix A — Literature Matrix

Include a comparison table such as the following (add one row per paper).

Paper	DSP Technique	AI Model	Dataset	Key Findings	Research Gap	Limitations
Nguyen et al., 2024	MFCC + bandpass filter	CNN-LSTM	RAVDESS	High accuracy on clean speech	Limited noise robustness	Weak on noisy input
...	...	...	...	...	...	...

Appendix B — AI Declaration Template

Item	Description
AI Tool	
Purpose	
Stage Used	
Human Verification	
Final Responsibility Statement	

Appendix C — Project Checklist

- ☐ Team Formation
- ☐ Topic Registration
- ☐ Research Problem
- ☐ Research Objectives
- ☐ Research Questions
- ☐ Literature Review
- ☐ Research Gap
- ☐ Hypothesis
- ☐ DSP Pipeline
- ☐ Feature Extraction
- ☐ AI Model
- ☐ Experimental Design
- ☐ Baseline Comparison
- ☐ Ablation Study
- ☐ Error Analysis
- ☐ Discussion
- ☐ Ethics Statement
- ☐ AI Declaration
- ☐ AI Reflection
- ☐ Final Report
- ☐ Source Code

- ☐ README
- ☐ Presentation

Appendix D — Frequently Asked Questions

Q: Can we use ChatGPT / Gemini / Claude?

A: Yes. Generative AI may be used to support learning and research (brainstorming, writing, summarizing literature, explaining concepts). All usage must be disclosed in the AI Declaration and the individual AI Reflection.

Q: Can we use GitHub Copilot?

A: Yes, for code assistance and debugging. You remain fully responsible for the correctness and integrity of the submitted code, and must declare its use.

Q: Can we use pretrained models?

A: Yes, provided you cite them, respect their licenses, and clearly state what was reused versus what you implemented or fine-tuned yourself.

Q: Can we use Kaggle or other public datasets?

A: Yes. Use public datasets (DEAP, RAVDESS, EMO-DB, PhysioNet, Kaggle, etc.), cite them correctly, and comply with their licensing terms.

Q: Can we collect our own datasets?

A: Yes. Self-recorded data requires informed consent, anonymization of personal information, and a description of the consent procedure in the Ethics Statement.

Q: Can we reuse code from GitHub?

A: Yes, if properly credited and license-compliant. Reusing code without attribution is an academic-integrity violation.

Q: How should AI usage be declared?

A: Complete the AI Declaration (Appendix B) in the report appendix and submit the individual AI Reflection. State the tools, purpose, stages used, and how outputs were verified.

Q: What happens if experiments cannot be reproduced?

A: Reproducibility is graded. If the submitted package cannot re-run the experiments (missing seeds, versions, data, or instructions), the reproducibility criterion is not met and the score is reduced accordingly.